Project WorkFlow & Architecture

Date	21 October 2026
Team ID	VAI-AI-2026
Project Name	VoyageAI — Personalized AI Travel Itinerary Planner
Maximum Marks	3 Marks

Project WorkFlow & Lifecycle Architecture:

The VoyageAI Project Workflow defines the end-to-end data pipeline and execution lifecycle—from traveler parameter capture to AI synthesis, database persistence, interactive timeline customization, and offline PDF export.

1. End-to-End System Workflow Stages:

Stage	Workflow Phase	Technical Action & Logic	Input / Trigger	Output / Result
01	**Preference Capture**	Traveler selects destination, travel dates, total budget, currency, style, and vibe tags.	TripBuilder 3-step UI	Validated trip configuration state
02	**Prompt Structuring**	FastAPI backend validates JWT, calculates per-day budget, and synthesizes structured JSON schema prompt.	POST /api/itinerary/generate	Engineered system & user prompts
03	**Groq LPU AI Synthesis**	High-speed LLM (openai/gpt-oss-120b) generates complete multi-day schedule with morning, afternoon, and evening slots.	Hardware JSON API call	Sub-second structured JSON itinerary
04	**Database Persistence**	Asyncpg transactional session persists trip, days, activities, and AI audit metrics into Supabase PostgreSQL.	SQLAlchemy 2.0 ORM	Relational records with RLS isolation
05	**Interactive Timeline**	Frontend renders chronological activities with time slots, location pins, cost counters, and edit/delete controls.	ItineraryView component	Responsive visual schedule
06	**Single-Day Re-roll**	AI re-synthesizes only one targeted day without altering the rest of the trip plan.	POST /api/itinerary/day/regenerate	Updated day schedule & DB sync
07	**Offline PDF Export**	Printer-friendly CSS formatting exports clean, offline-ready itineraries for travelers on the move.	Print Itinerary action	Clean printable travel guide

2. Milestone Development Lifecycle:

Milestones 1-2: Environment & Database	Git repository, Cursor AI agent, Supabase PostgreSQL with 7 relational tables & RLS policies.
Milestones 3-4: Recommendation & AI Engine	Preference capture, Groq LPU prompt engineering, JSON mode enforcement, and sub-second generation.
Milestones 5-7: Productization & Cloud Deployment	Interactive timeline, single-day regeneration, PDF export, Render backend & Vercel edge deployment.